

Econometrics I

Lecture 13c: Count Data

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- ▶ Dependent variable is a count: $y \in \{0, 1, 2, \dots\}$.
- ▶ Examples: purchases per customer per quarter, store visits, insurance claims, patents per firm, doctor visits, customer complaints, app sessions, number of children.
- ▶ Three things a linear model gets wrong: predictions go negative, the variance grows with the mean (heteroskedasticity by construction), and the pile of zeros is not a normal residual.
- ▶ Plan: Poisson as the baseline and why it is robust; overdispersion and the negative binomial; exposure; zero inflation and how to choose among models; counts in panels with fixed effects.
- ▶ Readings: Wooldridge Ch. 19; Cameron and Trivedi Ch. 20.

Poisson distribution

- **Poisson process**: events happen independently over time at rate λ per unit time. The **Poisson distribution** gives the distribution of the number of events.

- Probability mass:

$$\mathbb{P}(Y = y \mid \lambda) = \frac{e^{-\lambda} \lambda^y}{y!}$$

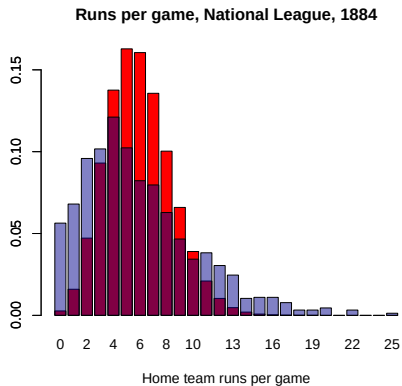
One parameter:

$$\mathbb{E}[y] = \lambda$$

$$\mathbb{V}[y] = \lambda$$

- **Equidispersion**, mean = variance, is a strong restriction. Real count data are almost always **overdispersed** (variance $>$ mean), which is the negative binomial's cue.

Poisson distribution



Plotted Poisson distribution (red) is for λ equal to the average runs per game.

Empirical probability mass function is in blue.

- Let the rate vary with observables:

$$\lambda(x) = \mathbb{E}[y \mid x] = \exp(x'\beta),$$

which keeps $\lambda(x) > 0$ and makes β_j a semi-elasticity.

- Log likelihood from the Poisson pmf:

$$\ell(\beta) = \sum_{i=1}^n \{-\exp(x_i'\beta) + y_i x_i'\beta - \ln y_i!\}$$

- First-order condition:

$$\sum_{i=1}^n (y_i - \exp(x_i'\beta)) x_i = 0.$$

- Globally concave: Newton's method converges from anywhere.

- Marginal effects:

$$\frac{\partial \mathbb{E}[y \mid x]}{\partial x_j} = \exp(x' \beta) \beta_j = \mathbb{E}[y \mid x] \beta_j$$

- Equivalently,

$$\frac{1}{\mathbb{E}[y \mid x]} \frac{\partial \mathbb{E}[y \mid x]}{\partial x_j} = \beta_j,$$

so β_j is the **proportional change in the mean** for a unit change in x_j . For a dummy, $e^{\beta_j} - 1$ is the exact proportional change.

- Compare the transformations deck (Week 2): this is the “log-linear” interpretation *without* taking logs of y , so zeros are not a problem. Hold that thought for the panel section.

Poisson is robust: consistency needs only the mean

- ▶ The first-order condition $\sum_i (y_i - \exp(x_i'\beta)) x_i = 0$ is a moment condition. It holds in the population whenever $\mathbb{E}[y_i | x_i] = \exp(x_i'\beta)$.
- ▶ So the Poisson MLE is consistent for β **even if the data are not Poisson**: overdispersed, underdispersed, zero-heavy, whatever. Only the conditional mean has to be right. (Gourieroux, Monfort and Trognon 1984: **pseudo-maximum likelihood**.)
- ▶ What breaks is the *variance*: the MLE standard errors assume $\mathbb{V}[y | x] = \mathbb{E}[y | x]$. With overdispersion they are too small, often by a lot.
- ▶ Fix: sandwich (robust) standard errors, Week 3. In `fixest`, `fepois` with `vcov = "hetero"` or a cluster.
- ▶ This is why Poisson regression has become the default for any non-negative outcome with zeros, including dollar amounts (Santos Silva and Tenreyro 2006, in trade; now everywhere).

- ▶ Counts are usually *per something*: purchases per **months active**, claims per **policy-year**, patents per **R&D dollar**, accidents per **miles driven**.
- ▶ If events arrive at rate $\lambda(x)$ per unit of exposure E_i , then
$$\mathbb{E}[y_i \mid x_i, E_i] = E_i \exp(x_i' \beta) = \exp(x_i' \beta + \ln E_i).$$
- ▶ So include $\ln E_i$ as a regressor with its coefficient **fixed at one**: an **offset**. Everything else is unchanged; β is now the effect on the *rate*.
- ▶ Estimating the coefficient on $\ln E_i$ instead of fixing it is a test of proportionality; fine as a check, but the offset is the model.
- ▶ The running example below has customers active for one, two, or three months of the quarter.

Overdispersion and the negative binomial

- Put unobserved heterogeneity in the rate:

$$y \mid \lambda, \nu \sim \text{Poisson}[\lambda\nu], \quad \nu \sim \text{Gamma}[\text{mean } 1, \text{variance } \alpha].$$

Integrating out ν : y is **negative binomial** with mean λ and variance $\lambda + \alpha\lambda^2$.

- Same mean as Poisson. Extra variance grows with λ^2 : heavy buyers are *much* more variable. $\alpha = 0$ is Poisson, so an LR test of $\alpha = 0$ is a test for overdispersion.
- The classic description (failures before r successes) is the same distribution with $r = 1/\alpha$ and $p = \lambda\alpha/(1 + \lambda\alpha)$; useful for simulation.
- **When to bother.** If you only want β in the conditional mean, Poisson with robust standard errors already has it. Estimate the NB when you need the *distribution*: predicting the share of customers with zero or ten-plus purchases, simulating, or pricing risk.

Running example: purchases per customer-quarter

- ▶ 4,000 customers. Covariates: **promo** exposure (half) and standardized **tenure**; exposure is months active in the quarter (1, 2, or 3).
- ▶ Ground truth, in three layers:

$$\lambda_i = E_i \exp(-0.5 + 0.6 \cdot \text{promo}_i + 0.2 \cdot \text{tenure}_i) \cdot \nu_i, \quad \nu_i \sim \text{Gamma}(\text{var } 0.8)$$

$y_i = 0$ with probability **0.25** (never buys this category), else $\text{Poisson}(\lambda_i)$.

- ▶ Sample: mean 1.3, variance 4.9, and 55% zeros. Overdispersed, zero-heavy: ordinary.
- ▶ Fit Poisson, negative binomial, and a zero-inflated negative binomial, and compare.

Zero inflation and hurdles

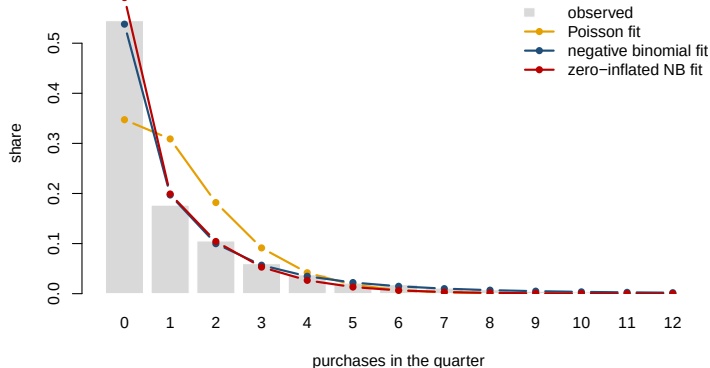
- ▶ Two reasons for a zero: the customer *could* have bought and did not (a Poisson zero), or is *not in the market* (a structural zero). With both, no single rate fits zeros and positives.
- ▶ **Zero-inflated** model: with probability π_i the count is zero for sure; otherwise it is Poisson or NB. The likelihood is a mixture:

$$\mathbb{P}(y = 0) = \pi + (1 - \pi) f(0 \mid \lambda), \quad \mathbb{P}(y = k) = (1 - \pi) f(k \mid \lambda), \quad k \geq 1.$$

Let π_i depend on covariates through a logit if “who is in the market” is itself the question.

- ▶ **Hurdle** model: a binary model for $y > 0$, then a *truncated* count model for $y \geq 1$; all zeros are structural and the two parts estimate separately.
- ▶ Neither is in `fixest`; the likelihood is ten lines in `optim` (workbook). Same idea as the Tobit deck’s two-part model: if the zeros are a different decision, give them their own equation.

Fit: observed versus predicted distributions



The Poisson fit has too few zeros and too few large counts, the signature of overdispersion. NB fixes the tail; the zero-inflated NB fixes the zeros as well. All three have nearly the same $\hat{\beta}$, which is the robustness point: the *mean* was never the problem.

Choosing among models: AIC, BIC, and Vuong

Three models with different likelihoods, not all nested. Tools:

- ▶ **Likelihood ratio** for nested pairs: Poisson within NB ($\alpha = 0$, on the boundary: halve the p -value).
- ▶ **Information criteria** for any pair: $AIC = -2\ell + 2k$, $BIC = -2\ell + k \ln n$. AIC estimates expected out-of-sample fit (Kullback–Leibler distance to the truth); BIC penalizes harder and picks the true model with probability $\rightarrow 1$ if it is a candidate. Lower is better.
- ▶ **Vuong (1989)** for non-nested pairs: with $m_i = \ln f_A(y_i) - \ln f_B(y_i)$,
 $z = \sqrt{n} \bar{m} / s_m \xrightarrow{d} N(0, 1)$ under H_0 that the two models are equally close to the truth. A t -test on pointwise log-likelihood differences; positive favors A ; neither model has to be true.
- ▶ None of these tests whether β is right. They rank *distributions*: use them for prediction and simulation, not to decide whether an effect exists.

Estimates on the simulated customers

	Poisson	Poisson, robust SE	Negative binomial	Zero-inflated NB
Promo (truth = 0.6 among buyers)	0.631 (0.029)	0.631 (0.051)	0.620 (0.051)	0.615 (0.049)
Tenure (truth = 0.2)	0.213 (0.014)	0.213 (0.026)	0.204 (0.026)	0.203 (0.024)
Overdispersion α (truth = 0.8)	0 (imposed)	—	1.64	0.85
Never-buyer share π (truth = 0.25)	—	—	—	0.24
Log-likelihood	-7200.5	(same)	-5827.1	-5812.9
AIC / BIC	14407 / 14426		11662 / 11687	11636 / 11667
Vuong, ZINB vs. NB			$z = 2.73$ (favors ZINB)	

- All four columns agree on the mean effects; the Poisson MLE standard errors are half the robust ones.
- NB alone blames the excess zeros on overdispersion ($\hat{\alpha} = 1.6$ vs. 0.8); the zero-inflated version separates the two, and AIC, BIC, and Vuong all prefer it.

Counts in panels: fixed effects work here

- ▶ Customers $\times$ quarters, with a customer effect η_i in the rate: $\mathbb{E}[y_{it} \mid x_{it}, \eta_i] = \exp(x'_{it}\beta + \eta_i)$.
- ▶ Unlike the logit (incidental parameters, Week 12), the Poisson fixed-effects MLE is **consistent for fixed T** : conditioning on $\sum_t y_{it}$ removes η_i exactly, and the conditional MLE coincides with the plain MLE with customer dummies (Hausman, Hall and Griliches 1984; Lancaster 2000). Same robustness: only the mean has to be right.
- ▶ Practical facts: customers with $y_{it} = 0$ in every period drop out (they carry no information about β ; `fepois` tells you how many). Cluster by customer.
- ▶ The alternative people reach for is OLS on $\ln(1 + y_{it})$ with customer FE. The transformations deck (Week 2) explained why $\ln(1 + y)$ has no units-invariant interpretation; here is what it does to a known effect.

Poisson pseudo-ML with fixed effects versus $\ln(1 + y)$

Simulated: 1,500 customers $\times$ 8 quarters. Heavy buyers (η_i high) are targeted with more promotions.
True promo effect: 0.6 log points on the rate.

	Poisson, no FE	Poisson + customer FE	OLS $\ln(1 + y)$ + FE	OLS $\ln y$ ($y > 0$)
Promo (truth = 0.6 log points)	-0.678 (0.027)	0.590 (0.024)	0.222 (0.010)	0.253 (0.015)
Observations	12000	11200	12000	5365
Share of zeros in y			0.54	

- ▶ Without FE the targeting flips the sign. With FE, Poisson recovers 0.6.
- ▶ $\ln(1 + y)$ with the same FE gets a third of the effect, and dropping the zeros does no better. Neither number is an elasticity of anything.

Everything except zero inflation is fixest:

```
fepois(y ~ promo + tenure, d, offset = ~log(months), vcov = "hetero") # Poisson, robust SE
fenegbin(y ~ promo + tenure, d, offset = ~log(months))                # negative binomial (+ theta)
fepois(y ~ promo | id, panel, vcov = ~id)                             # Poisson with customer FE
```

Zero-inflated NB by hand (the workbook fills in the details):

```
nll <- function(th) { la <- exp(X %*% th[1:3] + off); a <- exp(th[4]); pi <- plogis(th[5])
  f <- dnbinom(y, size = 1/a, mu = la)
  -sum(ifelse(y == 0, log(pi + (1 - pi) * f), log(1 - pi) + log(f))) }
optim(start, nll, method = "BFGS", hessian = TRUE)
```

► Python: `pf.fepois` in `pyfixest`; `statsmodels` has NB, zero-inflated, and hurdle models.

- ▶ Poisson regression is a model of the *conditional mean* with a log link. It is consistent whenever the mean is right, Poisson or not. Use robust standard errors.
- ▶ Exposure goes in as an offset. Overdispersion goes in as gamma heterogeneity (negative binomial) when you need the distribution, not just the mean.
- ▶ Excess zeros: ask whether they are a different decision. Zero-inflated and hurdle models give them their own equation. AIC, BIC, and Vuong rank the distributions; they do not test β .
- ▶ In panels, Poisson with unit fixed effects is consistent at fixed T and is the right replacement for $\ln(1 + y)$ regressions.

`inclass_session13.Rmd`:

1. Tobit: reproduce the estimates table; then add heteroskedasticity to ε and watch the probit check fail.
2. Durations: reproduce Kaplan–Meier and Cox; then build the person-month data and confirm the cloglog with month dummies gives the same $\hat{\beta}$.
3. Counts: reproduce the Poisson/NB/ZINB comparison and the panel table; then make promo exposure *independent* of η_i and see which columns change.